

Chapter 39

Equation of a Circle

$$(x - h)^2 + (y - k)^2 = r^2$$

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

1) What is the **center of the circle** with the equation

$$(x - 5)^2 + (y + 3)^2 = 36?$$

- A) $(-5, 6)$ B) $(6, -5)$ C) $(-5, 3)$ D) $(5, -3)$ E) $(6, -3)$

2) What is the **length of the diameter** of the circle with the equation

$$(x + 3)^2 + y^2 = 36$$

- A) 12 B) 18 C) 6 D) 3 E) 36

3) Which of the following is **the equation of a circle** with the center at the origin and a radius of 5 units?

- A) $x^2 + y^2 = 5$
B) $x^2 + y^2 = 25$
C) $x^2 - y^2 = 5$
D) $(x - 5)^2 + (y - 5)^2 = 25$
E) $(x - 5)^2 - (y - 5)^2 = 25$

4) Which of the following is **one of the x-intercepts** of a circle with center $(-3, -2)$ and a radius of 4?

- A) 0
B) $3 - 3\sqrt{2}$
C) $-3 + 2\sqrt{3}$
D) $2 + \sqrt{7}$
E) -2

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

5) What is the **radius of a circle** with the equation $(x + 5)^2 + (y + 3)^2 = 5$?

- A) $\sqrt{5}$ B) 5 C) $\sqrt{10}$ D) 10 E) $\sqrt{8}$

6) What is the **equation of a circle** with center (1, 4) and radius 3?

- A) $(x + 1)^2 + (y + 4)^2 = 9$
 B) $(x + 1)^2 - (y + 4)^2 = 3$
 C) $x^2 - y^2 = 3$
 D) $(x - 1)^2 + (y - 4)^2 = 9$
 E) $(x - 1)^2 - (y + 4)^2 = 9$

7) What is the **equation of a circle** with the center at the origin and a radius of 10?

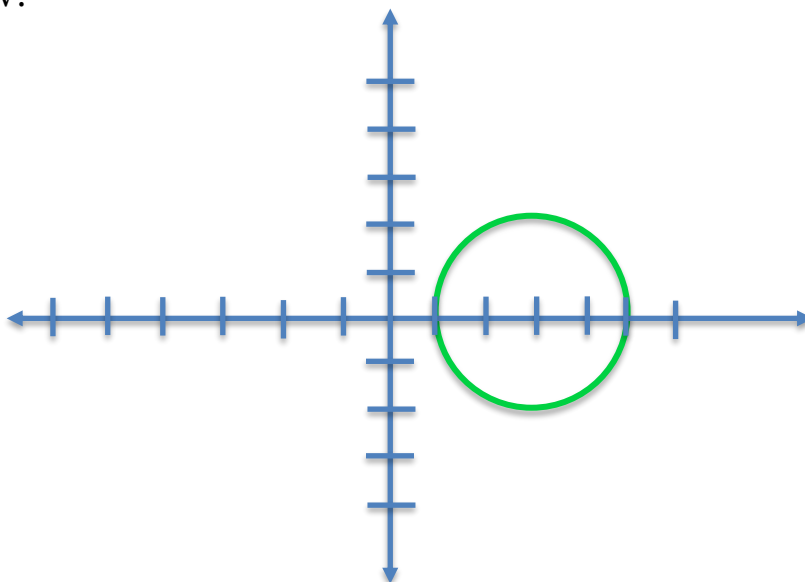
- A) $x^2 + y^2 = 100$
 B) $x^2 - y^2 = 100$
 C) $x^2 - y^2 = 10$
 D) $(x - 0)^2 + (y - 0)^2 = 10$
 E) $(x - 10)^2 + (y + 10)^2 = 0$

8) What is the **equation of a circle** tangent to both the x-axis and y-axis at 3?

- A) $x^2 + y^2 = 9$
 B) $x^2 + y^2 = 36$
 C) $(x + 6)^2 + (y + 6)^2 = 36$
 D) $(x + 3)^2 + (y + 3)^2 = 36$
 E) $(x - 3)^2 + (y - 3)^2 = 9$

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

9) Which of the following equations best represents the **circle** from the graph below?



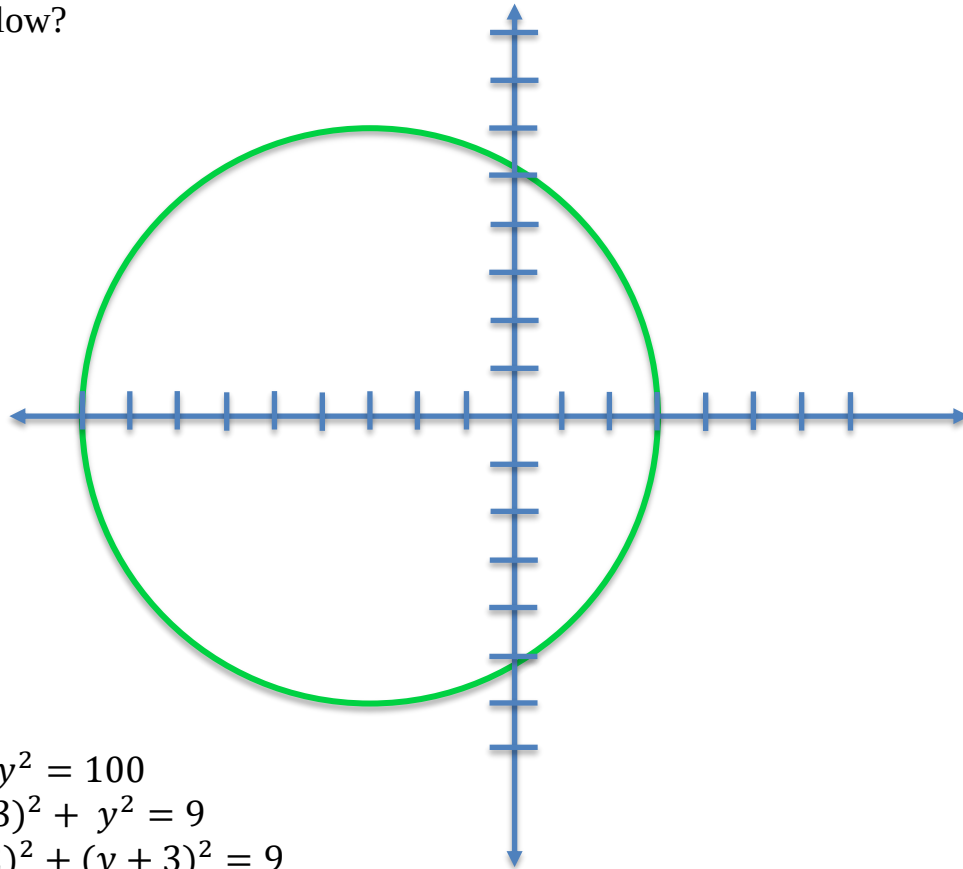
- A) $x^2 + y^2 = 4$
- B) $(x - 3)^2 + y^2 = 9$
- C) $(x - 3)^2 + (y + 3)^2 = 9$
- D) $(x + 3)^2 + y^2 = 4$
- E) $(x - 3)^2 + y^2 = 4$

10) What is the **center and radius** of the following circle;
 $(x + 9)^2 + (y - 9)^2 = 81$

- A) Center $(-9, 9)$ radius = 3
- B) Center $(9, -9)$ radius = 9
- C) Center $(-9, 9)$ radius = 81
- D) Center $(9, -9)$ radius = 81
- E) Center $(-9, 9)$ radius = 9

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

11) Which of the following equations best represents the circle from the graph below?



- A) $x^2 + y^2 = 100$
- B) $(x + 3)^2 + y^2 = 9$
- C) $(x - 2)^2 + (y + 3)^2 = 9$
- D) $(x - 3)^2 + y^2 = 9$
- E) $(x + 3)^2 + y^2 = 36$

12) Which of the following is the center and radius of a circle with the equation $x^2 + y^2 + 4x - 2y - 11 = 0$.

- A) Center $(4, -2)$; Radius = 5
- B) Center $(4, 1)$; Radius = 11
- C) Center $(-3, 5)$; Radius = 6
- D) Center $(-2, 1)$; Radius = 4
- E) Center $(5, -2)$; Radius = 3

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

13) Which of the following is the center and radius of a circle with the equation $x^2 + y^2 + 6x - 10y = 2$

- A) Center $(4, -2)$; Radius = 5
- B) Center $(4, 1)$; Radius = 11
- C) Center $(-3, 5)$; Radius = 6
- D) Center $(-2, 1)$; Radius = 4
- E) Center $(5, -2)$; Radius = 3

14) What is the longest distance for two points that lie on the circle $(x - 2)^2 + (y + 3)^2 = 9$?

- A) 3 units
- B) 5 units
- C) 7 units
- D) 6 units
- E) 8 units

Solutions Chapter 39

Equation of a Circle

- | | |
|------|-------|
| 1) D | 8) E |
| 2) A | 9) E |
| 3) B | 10) E |
| 4) C | 11) E |
| 5) A | 12) D |
| 6) D | 13) C |
| 7) A | 14) D |

1) The equation of a circle is;

$(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle $= r$. Since the equation of this circle is $(x - 5)^2 + (y + 3)^2 = 36$, the circle has its center at $(5, -3)$. Notice the signs in the equation are opposite of the actual center because the equation is $(x - h)$ and $(y - k)$. The equation can also be written as $(x - (5))^2 + (y - (-3))^2 = 36$ to better show the center.

$(5, -3)$ Answer D Difficulty Level 4

2) The length of the diameter of any circle is twice the length of its radius. Since the equation of a circle is given by $(x - h)^2 + (y - k)^2 = r^2$, the radius of $(x + 3)^2 + y^2 = 36$ is the square root of 36; $\sqrt{36} = 6$. Remember, the circle equation gives radius squared. To find the radius, take the square root of r^2 . Since the length of the radius is 6, the length of the diameter $= 2 \times 6 = 12$.

12 Answer A Difficulty Level 5

3) The equation of a circle $(x - h)^2 + (y - k)^2 = r^2$, with the center at the origin, $(0, 0)$; can be written as $(x - 0)^2 + (y - 0)^2 = r^2$;

$x^2 + y^2 = r^2$ since h and k are both zero;

Since the radius $= 5$, $r^2 = (5)^2 = 25$.

The equation of a circle centered at $(0, 0)$ with radius 5 is equal to $x^2 + y^2 = 25$.

$x^2 + y^2 = 25$ Answer B Difficulty Level 4.5

4) To find an x-intercept of the circle, start by making an equation of a circle from the given information. Since the circle has its center at $(-3, -2)$ and has a radius of 4, (The equation of the circle $(x - h)^2 + (y - k)^2 = r^2$, the center is (h, k) and the radius is r) Substitute for h, k and r ; $(x - (-3))^2 + (y - (-2))^2 = (4)^2$;
 $(x + 3)^2 + (y + 2)^2 = 16$. To find x-intercepts of any equation, set y equal to 0 and solve for x. $(x + 3)^2 + (0 + 2)^2 = 16$; $(x + 3)^2 = 12$; take the square root of both sides of the equation;
 $x + 3 = \pm\sqrt{12}$, subtract 3 from both sides $x = -3 \pm \sqrt{12} = -3 \pm 2\sqrt{3}$.
 Therefore, there are two solutions or two x-intercepts;
 $(-3 + 2\sqrt{3}, 0)$ and $(-3 - 2\sqrt{3}, 0)$. Both answer choices are correct, but only one is a possible answer choice.

$-3 + 2\sqrt{3}$ Answer C Difficulty Level 9

5) The equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$ with r is equal to the radius. To find the radius of the circle, take the square root of r^2 from the given equation; $(x + 5)^2 + (y + 3)^2 = 5$. Since $r^2 = 5$, solve for r (the radius) by taking the square root of both sides of the equation. $r = \sqrt{5}$.

$\sqrt{5}$ Answer A Difficulty Level 5

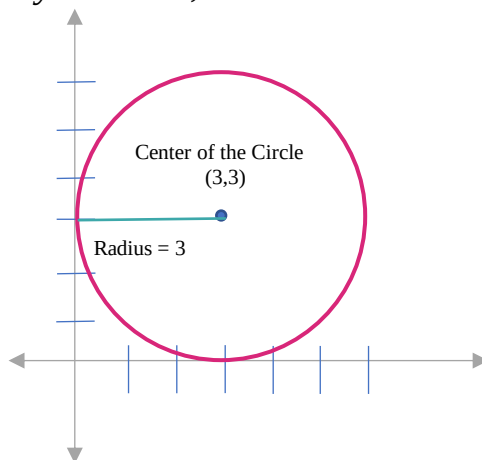
6) To find the equation of a circle $(x - h)^2 + (y - k)^2 = r^2$ substitute for the center; (h, k) and the radius, r . The center of the circle (h, k) is $(1, 4)$ and the circle has a radius of 3. Therefore, $h = 1$, $k = 4$, and $r = 3$. Substitute; $(x - (1))^2 + (y - (4))^2 = (3)^2$; $(x - 1)^2 + (y - 4)^2 = 9$.

$(x - 1)^2 + (y - 4)^2 = 9$ Answer D Difficulty Level 4.5

7) The equation of a circle $(x - h)^2 + (y - k)^2 = r^2$, with the center at the origin, $(0, 0)$; $(x - 0)^2 + (y - 0)^2 = r^2$; is written as $x^2 + y^2 = r^2$ since h and k are both zero. Since the radius(r) = 10, $r^2 = (10)^2 = 100$.

The equation of a circle centered at $(0, 0)$ with radius 10 is equal to $x^2 + y^2 = 100$.
 $x^2 + y^2 = 100$ Answer A Difficulty Level 4.5

8) To find the equation of a circle, find the center and radius of the circle. A circle tangent to both the x and y-axis at 3;



has its center at $(3, 3)$ and has a radius of length 3. Substitute the center and radius into the circle equation; $(x - h)^2 + (y - k)^2 = r^2$. (Remember, the center of the circle is (h, k) and the radius of the circle $= (r)$); $(x - 3)^2 + (y - 3)^2 = 3^2$

$(x - 3)^2 + (y - 3)^2 = 9$ Answer E Difficulty Level 7

9) To find the equation of a circle $(x - h)^2 + (y - k)^2 = r^2$ substitute for the center; (h, k) and the radius, r . The center of the circle (h, k) is $(3, 0)$ and the circle has a radius of 2. Therefore, $h = 3$, $k = 0$, and $r = 2$.

Substitute; $(x - (3))^2 + (y - (0))^2 = (2)^2$; $(x - 3)^2 + y^2 = 4$.

$(x - 3)^2 + y^2 = 4$ Answer E Difficulty Level 5

10) The equation of a circle is; $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle $= r$. Since the equation of this circle is $(x + 9)^2 + (y - 9)^2 = 81$, the circle has its center at $(-9, 9)$. Notice the signs in the equation are opposite of the actual center because the equation is $(x - h)$ and $(y - k)$. The equation can also be written as $(x - (-9))^2 + (y - (9))^2 = 81$ to better show the center. To find the length of the circle's radius(r), take the square root of r^2 . Since $r^2 = 81$, $r = 9$.

Center $(-9, 9)$ Radius $= 9$ Answer E Difficulty Level 5

11) To find the equation of a circle $(x - h)^2 + (y - k)^2 = r^2$ substitute for the center; (h, k) and the radius, r . The center of the circle (h, k) is $(-3, 0)$ and the circle has a radius of 6. Therefore, $h = -3$, $k = 0$, and $r = 6$.

Substitute; $(x - (-3))^2 + (y - (0))^2 = (6)^2$; $(x + 3)^2 + y^2 = 36$.

$(x + 3)^2 + y^2 = 36$ Answer E Difficulty Level 5

12) To find the center and radius of the circle, transform the given equation into an equation of a circle by completing the square. To complete the square, start by rewriting the equation in this form;

$ax^2 + bx + \underline{\hspace{1cm}} + dy^2 + fy + \underline{\hspace{1cm}} = C$, where C is a constant. Then, take half of b , square it, place it in the first space, then add it to the right side of the equation. Next, take half of f , square it, place it in the second space, and then add it to the right side of the equation. Finally, factor the x's and the y's separately on the left-hand side of the equation and sum the right side of the equation.

Set up the equation to complete the square by adding 11 to both sides of the equation and separating the x and y variables on the left-hand side of the equation;

$x^2 + 4x + \underline{\hspace{1cm}} + y^2 - 2y + \underline{\hspace{1cm}} = 11$. Take half of the middle term from the x equation and square it. Place this number in the space and add it to the right-hand side of the equation.

For the x-variables, take half of 4; $4/2 = 2$ and square it $2^2 = 4$. Add 4 to the first space, then add 4 to the right-hand side of the equation. For the y-variables, take half of -2; $-2/2 = -1$, then square it $(-1)^2 = 1$. Add 1 to the second space, then add 1 to the right-hand side of the equation

$x^2 + 4x + 4 + y^2 - 2y + 1 = 11 + 4 + 1$. Finally, factor both perfect square trinomials and sum the right side of the equation; $(x + 2)^2 + (y - 1)^2 = 16$

Center $(-2, 1)$ Radius = 4 Answer D Difficulty Level 9

13) To find the center and radius of the circle, transform the given equation into an equation of a circle by completing the square. To complete the square, start by rewriting the equation in this form;

$ax^2 + bx + \underline{\hspace{1cm}} + dy^2 + fy + \underline{\hspace{1cm}} = C$, where C is a constant. Then, take half of b , square it, place it in the first space, then add it to the right side of the equation. Next, take half of f , square it, place it in the second space, and then add it to the right side of the equation. Finally, factor the x's and the y's separately on the left-hand side of the equation and sum the right side of the equation.

Set up the equation to complete the square by separating the x and y variables on the left-hand side of the equation;

$x^2 + 6x + \underline{\hspace{1cm}} + y^2 - 10y + \underline{\hspace{1cm}} = 2$. Take half of the middle term from the x equation and square it. Place this number in the space and add it to the right-hand side of the equation.

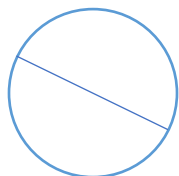
For the x -variables; take half of 6 ; $6/2 = 3$ and square it $(3)^2 = 9$. Add 9 to the first space, then add 9 to the right-hand side of the equation. For the y -variables, take half of -10 ; $-10/2 = -5$, then square it $(-5)^2 = 25$. Add 25 to the second space, then add 25 to the right-hand side of the equation

$$x^2 + 6x + 9 + y^2 - 10y + 25 = 2 + 9 + 25.$$

Finally, factor both perfect square trinomials and sum the right side of the equation; $(x + 3)^2 + (y - 5)^2 = 36$; The equation of a circle is; $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle $= r$. Since the equation of this circle is $(x + 3)^2 + (y - 5)^2 = 36$, the circle has its center at $(-3, 5)$, and the length of the radius is $\sqrt{r^2} = \sqrt{36} = 6$

Center $(-3, 5)$ Radius $= 6$ Answer C Difficulty Level 9

14) The longest distance between two points that lie on a circle is the diameter of the circle.



Since the equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where r is the length of the radius and the diameter is twice as long as the radius. Since the given equation of the circle; $(x - 2)^2 + (y + 3)^2 = 9$, $r^2 = 9$, $r = 3$. If the radius has a length of 3, the diameter has a length $(2 \times 3) = 6$

6 units Answer D Difficulty Level 6